

Introduction

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

What is Computational Intelligence (CI)?

Bezdek defines a computationally intelligent system such that

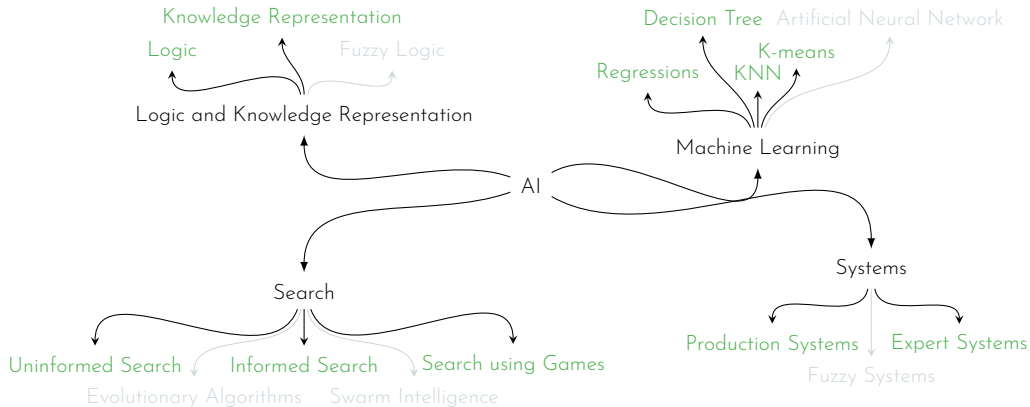
A system is computationally intelligent when it: deals with only numerical (low-level) data, has pattern recognition components, does not use knowledge in the AI sense; and additionally when it (begins to) exhibit (1) computational adaptivity; (2) computational fault tolerance; (3) speed approaching human-like turnaround, and (4) error rates that approximate human performance.

Computational intelligence focuses on biologically and linguistically motivated computational paradigms.

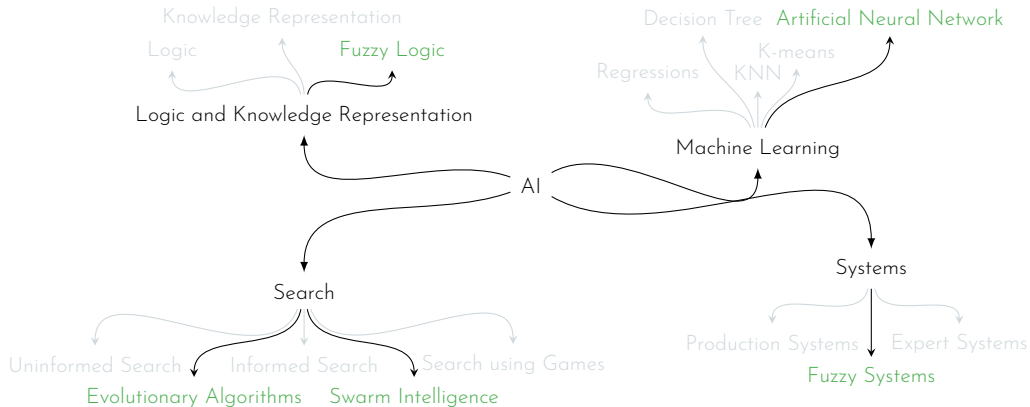
The primary paradigms in CI include **evolutionary computation (EC)**, **artificial neural network (ANN or NN)**, and **fuzzy systems (FS)**.

There are other paradigms in CI such as artificial immune systems (AIS) and artificial life (AL).

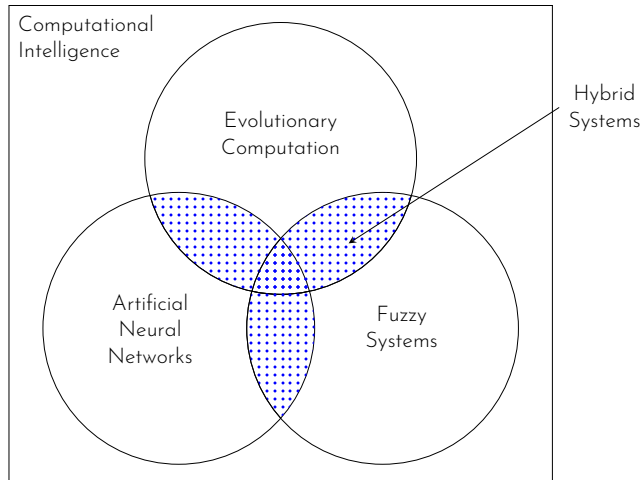
Topics covered in CSC3206 Artificial Intelligence (AI)



Topics covered in CSC3034 Computational Intelligence (CI)



Summary



Where are CI methods implemented in?

Evolutionary computation

- ▶ data mining
- ▶ classification
- ▶ clustering
- ▶ scheduling
- ▶ approximation
- ▶ route optimisation ...

Artificial neural networks

- ▶ diagnosis of diseases
- ▶ speech recognition
- ▶ data mining
- ▶ image processing
- ▶ pattern recognition
- ▶ planning game strategies ...

Fuzzy systems

- ▶ control systems
- ▶ gear transmission
- ▶ home appliances
- ▶ traffic signal control ...

Traditional set theory and binary-valued logic deal with elements or parameters in a black and white approach.

However, human reasoning is almost always not the case. In human language, it is common to use “some”, “most”, etc.

Fuzzy sets and fuzzy logic thus account for this type of reasoning, referred to as **approximate reasoning**. In fuzzy sets, an element belongs to a set to a certain **degree of certainty**. Fuzzy logic allows reasoning with these uncertain facts to infer new facts, with a degree of certainty associated with each fact.

Nonstatistical uncertainty (fuzziness)

Uncertainty in fuzzy systems based on *vagueness*, *impression*, and/or *ambiguity*. It is an inherent property of a system and cannot be changed or resolved by observations.

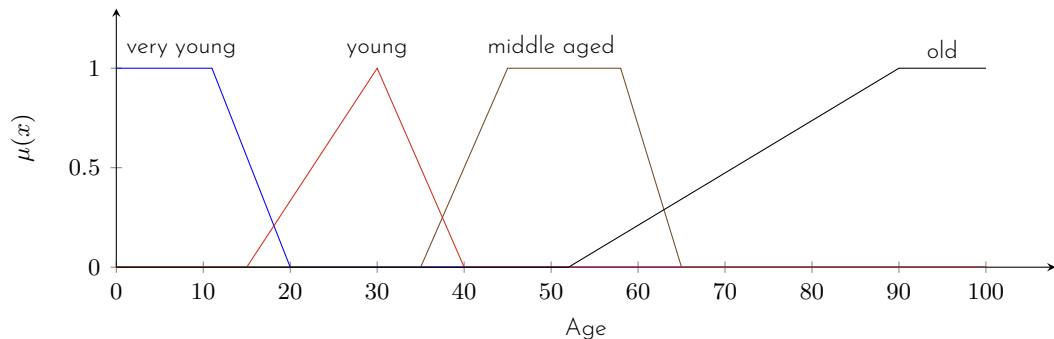
For example, Peter belongs to a fuzzy set of tall people with a degree of 0.9. Suppose the event is to determine if Peter is good at basketball. After the degree of membership for Peter to belong to the set of good basketball players is identified, Peter still belongs to the set of tall people with a degree of 0.9.

Statistical uncertainty

Uncertainty based on the laws of probability. The uncertainty is resolved through observations.

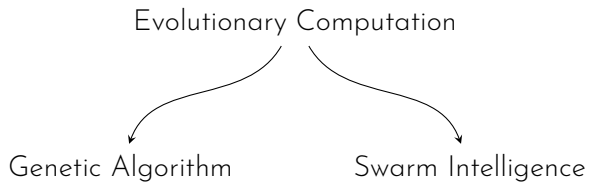
For example, before tossing a fair coin, there is a 50% probability to get a head and 50% probability to get a tail. After the event of tossing the coin, there is no uncertainty as to whether we will get a head or a tail.

Example of a fuzzy system¹:



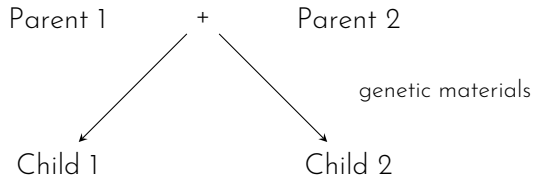
¹Example adapted from "Computational Intelligence: An Introduction" by Engelbrecht (2007)

Evolutionary computation



Genetic algorithm

Genetic algorithm aims to mimic the processes of natural evolution, i.e. survival of the fittest.



If an individual inherits bad characteristics → **weak and will not survive**

If an individual inherits good characteristics → **survive and reproduce**

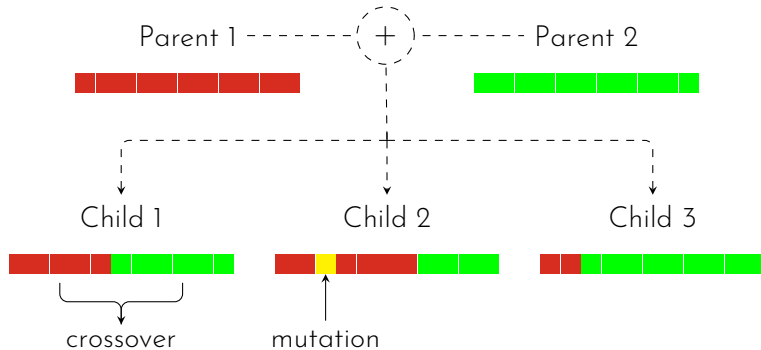
After multiple generations, the bad characteristics will be phased out while the good characteristics remain.

Two important processes: **crossover** and **mutation**.

crossover occurs during sexual reproduction, i.e. with >1 parents.

mutation (ideally) leads to the improvement of the individuals, i.e. more good characteristics.

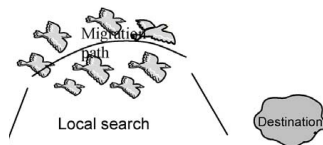
Evolutionary computation



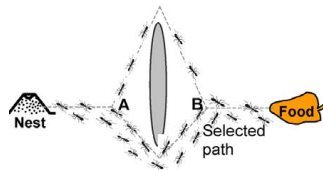
Swarm intelligence originates from the study of swarms of **social organisms** such as birds, ants, bees, etc.

The observation of social organisms suggests that intelligent group behavior can be achieved through the **simple interaction between individuals**. This observation has given rise to the design of very efficient optimisation and clustering algorithms.

Particle swarm optimisation (PSO) is modeled on the social behaviour of bird flocks.²



Ant colony optimisation (ACO) is modeled on the foraging behaviour of ants.¹



²Images extracted from "Comparison among five evolutionary-based optimization algorithms" by *Elbeltagi et al.* (2005)

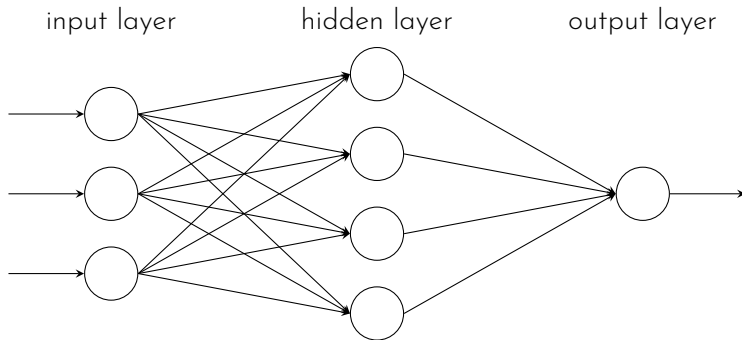
Artificial neural networks aims to mimic the biological neural systems.

The basic building blocks of biological neural systems are neurons that are interconnected through synapses.

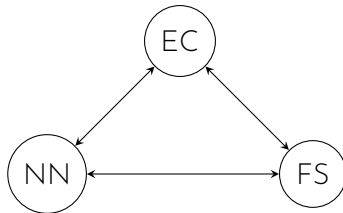
When we read, write, watch, smell, eat, live our daily life, the synapses between different neurons strengthen or weaken. Therefore, when I say “food”, the immediate impressions that come into your minds are different.

Artificial neural networks

Typical structure of an artificial neural network



Hybrid systems are formed by combining the techniques from different paradigms.



Progression and projection on AI

The Dartmouth College Artificial Intelligence Conference

Read through AI@50 and discuss

- ▶ Reflections on 1956
- ▶ Evaluations at 2006
- ▶ Projections to 2056

CSC3034 Current Course Update

01 Introduction - handout

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current labs connect core CI algorithms with physical-AI simulation examples.
- Isaac Sim examples use its bundled Python 3.12 environment and current launchers.
- Every simulator-dependent example includes a CPU/Matplotlib learning fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.